A Spatio-Temporal Foundation Model for Multi-Pollutant Air Quality Forecasting

Design specification, training protocol and evaluation plan

Design document

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Abstract

We specify a spatio-temporal foundation model (AQFM) that consumes irregularly sampled, block-missing, multi-station air quality observations and emits calibrated probabilistic forecasts for multiple pollutants simultaneously. The design is driven by three empirical findings from the Delhi CPCB exploratory analysis: (i) missingness is *systemic*, with $\mathbb{P}(\mathrm{PM}_{10}$ missing $\mid$ $\mathrm{PM}_{2.5}$ missing$) = 0.928$, so gaps are station-wide contiguous blocks rather than isolated dropouts; (ii) strong cross-pollutant structure exists ($r_{\mathrm{PM}_{2.5},\mathrm{PM}_{10}} = 0.84$, $r_{\mathrm{NO},\mathrm{NO}_x} = 0.90$) which can be imposed by construction instead of learned; and (iii) the 2018–2026 subset achieves 7–10% missingness on primary pollutants, making it the viable pre-training window. We compare four decoder designs, recommend a query-based unified decoder, and give a complete training and evaluation protocol.

Contents

1 Problem statement and notation **2**

2 Data **3**
2.1 Temporal scope . . . 3
2.2 Scale, and why Delhi alone is not enough . . . 3
2.3 Exogenous covariates . . . 3
2.4 Pre-processing . . . 3

3 Tokenisation **4**
3.1 One token per (station, channel, patch) . . . 4
3.2 Three tiers of irregularity . . . 4
3.3 Continuous time features . . . 4

4 Encoder **5**
4.1 Axial factorisation . . . 5
4.2 Masked attention . . . 5
4.3 Wind-aware spatial bias . . . 5
4.4 Per-station latent bottleneck . . . 5

5 Decoder: four designs **5**
5.1 Design A — one decoder per pollutant . . . 6
5.2 Design B — query-based unified decoder (recommended) . . . 6
5.3 Design C — shared trunk with per-pollutant adapters . . . 6
5.4 Design D — mixture-of-experts decoder . . . 6

6 Physical constraints **7**

7 Predictive distribution and loss **7**
7.1 Why not MSE . . . 7
7.2 Mixture head . . . 7
7.3 CRPS via the order-statistic identity . . . 7
7.4 Total objective . . . 8

8 Pre-training curriculum **8**
8.1 Mixed masking . . . 8
8.2 Two-stage schedule . . . 8

9 Reference configuration **9**

10 Evaluation protocol **9**
10.1 Metrics that matter . . . 9
10.2 Baselines . . . 9
10.3 Splits and leakage discipline . . . 9

11 Planned ablations **10**

12 Risks and honest caveats **10**

13 Implementation map **11**

1 Problem statement and notation

Let $\mathcal{S} = \{1, \dots, S\}$ index monitoring stations and $\mathcal{C} = \{1, \dots, C\}$ index channels (pollutants, meteorology, and derived quantities). Observations arrive on a nominal 15-minute grid but with substantial missingness. Write

$$X \in \mathbb{R}^{S \times C \times T}, \quad \text{values (log-transformed, see §2.4)} \tag{1}$$
$$M \in \{0,1\}^{S \times C \times T}, \quad \text{observation indicator, } M_{s,c,t} = 1 \text{ iff measured} \tag{2}$$
$$\tau \in \mathbb{R}^{T}, \quad \text{absolute timestamps in hours} \tag{3}$$
$$G \in \mathbb{R}^{S \times 2}, \quad \text{station coordinates (lat, lon)} \tag{4}$$
$$U \in \mathbb{R}^{S \times D_{\text{st}}}, \quad \text{static station covariates.} \tag{5}$$

Given a context window of length T_{ctx} and known-future covariates, the task is to produce the joint predictive distribution

$$p\big(X_{s,c,t} \;:\; s \in \mathcal{S},\; c \in \mathcal{C}_{\text{tgt}},\; t \in (T_{\text{ctx}}, T_{\text{ctx}} + H] \;\big|\; X_{\cdot,\cdot,\le T_{\text{ctx}}},\; M,\; \tau,\; G,\; U \big), \tag{6}$$

with H up to 72 hours. We require the model to be well-defined when (a) arbitrary channels are missing, (b) arbitrary contiguous blocks are missing across all channels at a station, and (c) the target set $\mathcal{C}_{\text{tgt}}$ and horizon set are chosen at inference time.

Design principle. Every irregularity is handled by *excluding tokens*, never by imputing values before the model sees them. Pre-imputation injects a second model's errors into the training signal and makes the observation mask unrecoverable.

2 Data

2.1 Temporal scope

The EDA is unambiguous: pre-2018 station-years show 80–100% missingness (PM_{10} is 100% missing every year through 2014), while 2018–2026 sits at 7–21% for primary pollutants. We therefore train on **2018–2026** and treat pre-2018 as an optional low-weight auxiliary corpus, not as pre-training data.

2.2 Scale, and why Delhi alone is not enough

Delhi's 38 stations yield $\approx 1.44 \times 10^7$ rows and $\approx 2.16 \times 10^8$ scalar observations. Time-series scaling laws differ from those for language, but the order of magnitude is informative: at the commonly cited $\sim$20 tokens per parameter this supports a model of roughly 10–30M parameters. The architecture below wants more.

The resolution is the national network: roughly 550 CPCB stations across India (Figure **??** in the accompanying analysis) give 10–15$\times$ the data. We therefore adopt a two-corpus strategy.

Corpus	Stations	Approx. observations	Role
All-India CPCB, 2018–2026	$\sim$550	$\sim 3 \times 10^9$	Stage-1 pre-training
Delhi NCR, 2018–2026	38	2.2×10^8	Stage-2 fine-tuning
Delhi NCR, pre-2018	38	–	Optional, low weight

Table 1: Two-corpus training strategy. National pre-training also supplies a genuine domain-shift test set: Chennai's coastal sea-breeze regime versus Delhi's continental winter inversion.

2.3 Exogenous covariates

Two additions are likely to matter more than any architectural choice.

1. **Boundary layer height.** Beyond roughly 12 hours of lead, forecast skill is dominated by ventilation rather than by emissions. The CPCB variable list contains no mixing-depth proxy. Obtain planetary boundary layer height, 10 m winds, and precipitation from ERA5 reanalysis.
2. **Known-future meteorology.** Numerical weather prediction output is available at forecast time. Train with ERA5 as a perfect-prognosis oracle, then substitute operational NWP at inference; report the skill gap between the two, since it bounds the achievable operational performance.

The satellite/reanalysis annual-mean $PM_{2.5}$ field supplies static per-station covariates U: climatological burden, urban–rural gradient, and a spatial prior for stations with short records.

2.4 Pre-processing

Concentrations are right-skewed and heavy-tailed, so all pollutant channels are mapped through $x \mapsto \log(1 + x)$ before any statistics are computed. Values are then normalised per (station, channel) using *context-only* statistics:

$$\hat{\mu}_{s,c} = \frac{\sum_{t \le T_{\text{ctx}}} M_{s,c,t} X_{s,c,t}}{\sum_{t \le T_{\text{ctx}}} M_{s,c,t}}, \qquad \hat{\sigma}^2_{s,c} = \frac{\sum_{t \le T_{\text{ctx}}} M_{s,c,t} (X_{s,c,t} - \hat{\mu}_{s,c})^2}{\sum_{t \le T_{\text{ctx}}} M_{s,c,t}}. \tag{7}$$

Statistics computed over the full window would leak the scale of the future into the input. When $\sum_t M_{s,c,t}$ falls below a threshold the transform falls back to the identity rather than dividing by a noise estimate.

3 Tokenisation

3.1 One token per (station, channel, patch)

Let P be the patch length ($P = 16$, i.e. 4 hours) and $N = T_{\text{ctx}}/P$. Define patch slices $x_{s,c}^{(n)} \in \mathbb{R}^P$ and $m_{s,c}^{(n)} \in \{0,1\}^P$. The token is

$$\boxed{z_{s,c,n} \;=\; W_v \begin{bmatrix} x_{s,c}^{(n)} \\ m_{s,c}^{(n)} \end{bmatrix} \;+\; e_c^{\text{ch}} \;+\; \phi_{\text{st}}\big([G_s; U_s]\big) \;+\; \psi_{\text{t}}\Big(\bar{\tau}^{(n)}\Big) \;\in\; \mathbb{R}^d} \tag{8}$$

with $W_v \in \mathbb{R}^{d \times 2P}$ and $\bar{\tau}^{(n)}$ the mean timestamp of patch n. Three properties follow directly from this construction.

Channel-count flexibility.
The token carries its own channel identity, so a station lacking NH_3 simply contributes fewer tokens. No padding, no sentinel values, no per-station bookkeeping.

Station generalisation.
ϕ_{st} is a function of coordinates, not a lookup table, so a station never seen in training is a valid input.

Mask visibility.
The mask enters the embedding alongside the values, so a patch that is 30% observed is not confused with a clean one.

3.2 Three tiers of irregularity

Irregularity	Mechanism	Rationale
Partial patch (some steps missing)	Mask channel in Eq. (8), zero-fill values	The embedding learns to discount unobserved positions
Fully unobserved patch	Token excluded from every attention key set	Attending to an all-zero token is worse than not attending
Non-uniform timestamps	ψ_{t} is a function of τ, not of position index	A 10-hour gap must look like 10 hours, not one step
Asynchronous sources (future)	Continuous-time front end (mTAN, or set attention over (t, v, c) triplets)	Only needed when fusing satellite overpasses or low-cost sensors

Table 2: Missingness is handled at three levels; the fourth row is a planned extension, not part of the base model.

3.3 Continuous time features

With log-spaced periods $\{p_k\}_{k=1}^{K_f}$ from 3 hours to one year and $\omega_k = 2\pi/p_k$,

$$\psi_{\text{t}}(\tau) = W_\tau \Big[\, \{\sin \omega_k \tau,\ \cos \omega_k \tau\}_{k=1}^{K_f},\ \text{cyc}_{24}(\tau),\ \text{cyc}_{168}(\tau),\ \text{cyc}_{8760}(\tau),\ \log(1 + \Delta\tau) \,\Big], \tag{9}$$

where $\text{cyc}_p(\tau) = (\sin 2\pi\tau/p,\ \cos 2\pi\tau/p)$ and $\Delta\tau$ is the gap to the previous patch. The explicit gap feature is what lets the model distinguish "next patch" from "next patch after a ten-hour outage."

4 Encoder

4.1 Axial factorisation

A single sample contains $S \times C \times N$ tokens; at $S = 12$, $C = 19$, $N = 32$ this is 7,296 tokens, and at the full 38-station Delhi network it exceeds 23,000. Full self-attention is $O(L^2)$ and out of reach. We factorise each encoder block into three cheap passes:

$$Z \leftarrow Z + \mathrm{Attn}_{\mathrm{time}}\big(\mathrm{LN}(Z)\big) \qquad \text{over } n,\ \text{batched over } (s,c) \tag{10}$$

$$Z \leftarrow Z + \mathrm{Attn}_{\mathrm{var}}\big(\mathrm{LN}(Z)\big) \qquad \text{over } c,\ \text{batched over } (s,n) \tag{11}$$

$$Z \leftarrow Z + \mathrm{Attn}_{\mathrm{space}}\big(\mathrm{LN}(Z)\big) \qquad \text{over } s,\ \text{batched over } (c,n) \tag{12}$$

$$Z \leftarrow Z + \mathrm{FFN}\big(\mathrm{LN}(Z)\big). \tag{13}$$

Cost drops from $O\big((SCN)^2\big)$ to $O\big(SCN(S+C+N)\big)$: at the figures above, a factor of roughly 300. Stacking $B_{\mathrm{ax}} = 4$ such blocks gives every token an effective receptive field over the whole grid.

4.2 Masked attention

Each attention uses a key-validity mask $v_{s,c,n} = \mathbb{1}[\sum_t m^{(n)}_{s,c,t} > 0]$, implemented as an additive $-\infty$ bias. If every key in a row is invalid the row is unmasked rather than producing NaN; the mask channel already tells the model the content is uninformative.

4.3 Wind-aware spatial bias

A station 8 km *upwind* is far more informative than one 8 km downwind. A distance-only graph cannot express this asymmetry. Let d_{ij} be the great-circle distance and θ_{ij} the bearing from station i to j, and let $\mathrm{WD}_i^{(n)}$, $\mathrm{WS}_i^{(n)}$ be the patch-mean wind direction and speed at the query station. The additive attention bias is

$$\boxed{b_{ij}^{(n)} = \mathrm{MLP}\Big(\log(1+d_{ij}),\ \cos\big(\theta_{ij} - \mathrm{WD}_i^{(n)}\big),\ \sin\big(\theta_{ij} - \mathrm{WD}_i^{(n)}\big),\ \mathrm{WS}_i^{(n)},\ \cos(\cdot)\,\mathrm{WS}_i^{(n)},\ e^{-d_{ij}/\lambda}\Big)} \tag{14}$$

emitted per attention head and added to the spatial attention logits. This is cheap, and the learned bias is inspectable: plotting which stations the model attends to under different wind regimes is a publishable result in itself.

4.4 Per-station latent bottleneck

Cross-attending $\sim$1.7k decoder queries against $\sim$7k encoder tokens is wasteful. Each station's $(C \times N)$ grid is compressed by a Perceiver-style cross-attention into $L = 16$ latents, giving $S \cdot L$ memory tokens. This preserves the spatial structure the wind bias operates on while cutting decoder cross-attention cost by roughly 30×.

5 Decoder: four designs

This is the central design question. All four options share the encoder above.

5.1 Design A — one decoder per pollutant

Each target pollutant p gets its own decoder $\mathcal{D}_p$ cross-attending to the shared memory.

For. Pollutants genuinely differ in dynamics. The EDA correlation matrix makes this concrete: ozone is the only pollutant *positively* correlated with temperature ($r = +0.34$) while $PM_{2.5}$ sits at $r = -0.49$, reflecting photochemical production versus accumulation under winter inversion. Separate decoders also allow adding a pollutant without touching the trunk, and independent per-pollutant loss weighting.

Against. Decoder parameters scale linearly in $|\mathcal{C}_{\text{tgt}}|$; the substantial structure that *is* shared across pollutants cannot be shared; and independent heads produce physically incoherent joint output — nothing prevents $\widehat{\text{PM}}_{2.5} > \widehat{\text{PM}}_{10}$.

5.2 Design B — query-based unified decoder (recommended)

One decoder. A query is constructed per prediction target:

$$q_{s,c,h} \;=\; e_c^{\text{tgt}} \;+\; e_h^{\text{lead}} \;+\; \phi_{\text{st}}([G_s; U_s]) \;+\; \psi_{\text{t}}\Big(\tau_h^{\text{fut}}\Big)\,, \tag{15}$$

followed by L_{dec} layers of {query self-attention, cross-attention to memory, FFN}. Specialisation lives in the query embedding rather than in separate weights.

For. Arbitrary target subsets at inference (forecast $PM_{2.5}$ alone with no wasted compute); arbitrary horizons, including untrained ones if lead time is encoded continuously; cross-pollutant coherence, because query self-attention lets the $PM_{2.5}$ prediction attend to the PM_{10} prediction before either is emitted; one weight set to scale when moving from 38 to 550 stations.

Against. Potential under-fitting of a pollutant with genuinely divergent dynamics — which is what Design C addresses.

5.3 Design C — shared trunk with per-pollutant adapters

Shared decoder trunk plus FiLM conditioning or low-rank (LoRA-style) adapters per pollutant. Roughly 2% additional parameters buys most of Design A's specialisation. *Deploy this only if ablation shows Design B under-fits a specific pollutant*, most plausibly O_3.

5.4 Design D — mixture-of-experts decoder

A router over k experts. Pollutants cluster naturally into families (particulates, NO_x, photochemical, meteorological); MoE lets that clustering be learned rather than imposed. Worth the complexity only at national scale with many channels.

	A: per-pollutant	B: query	C: adapters	D: MoE
Decoder params vs. $\lvert\mathcal{C}_{\text{tgt}}\rvert$	linear	constant	constant+ϵ	constant
Arbitrary target subset at inference	no	yes	yes	yes
Arbitrary / unseen horizon	no	yes	yes	yes
Joint cross-pollutant coherence	no	yes	yes	yes
Per-pollutant specialisation	high	medium	high	high
Implementation complexity	low	low	medium	high
Verdict	baseline	**recommended**	fallback	at scale

Table 3: Decoder design comparison. Design B is the recommendation; A is worth implementing as an ablation baseline because it is the intuitive choice and should be shown to be dominated.

6 Physical constraints

Independent heads can emit $\mathrm{PM}_{2.5} > \mathrm{PM}_{10}$ and $\mathrm{NO} + \mathrm{NO}_2 \neq \mathrm{NO}_x$. Both are free to fix by predicting a base quantity and a bounded fraction:

$$\boxed{\mathrm{PM}_{2.5} = f_{\mathrm{pm}} \cdot \mathrm{PM}_{10}, \quad f_{\mathrm{pm}} = \sigma(\tilde{f}_{\mathrm{pm}}) \in (0,1); \qquad \mathrm{NO} = f_{\mathrm{nox}} \cdot \mathrm{NO}_x, \quad \mathrm{NO}_2 = (1 - f_{\mathrm{nox}}) \cdot \mathrm{NO}_x.} \tag{16}$$

The model predicts $\{\log(1+\mathrm{PM}_{10}),\ \tilde{f}_{\mathrm{pm}},\ \log(1+\mathrm{NO}_x),\ \tilde{f}_{\mathrm{nox}}, \dots\}$; observables are reconstructed afterwards. Two implementation notes:

- The logit-fractions are computed in the data pipeline and appended as extra input channels, so they receive their own normalisation statistics and are predicted like any other channel.
- A fraction is treated as observed only when *both* constituents are observed and the denominator is non-trivial. Otherwise a missing PM_{10} silently manufactures a $\mathrm{PM}_{2.5}$ fraction.

This encodes $r_{\mathrm{PM}_{2.5},\mathrm{PM}_{10}} = 0.84$ and $r_{\mathrm{NO},\mathrm{NO}_x} = 0.90$ by construction rather than asking the model to rediscover them, and it eliminates an entire class of nonsense output at zero parameter cost.

7 Predictive distribution and loss

7.1 Why not MSE

Air quality decisions are threshold decisions: AQI bands, GRAP stages, school closures. A calibrated upper tail matters more than a good conditional mean, and an MSE-trained point head systematically under-predicts exactly the November–December episodes the forecast exists to warn about.

7.2 Mixture head

Each query emits, for each of P timesteps, a K-component Gaussian mixture in normalised $\log(1+x)$ space — equivalently a log-normal mixture in concentration space:

$$p(y) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}\big(y;\ \mu_k,\ \sigma_k^2\big), \qquad \pi = \mathrm{softmax}(\ell), \quad \log \sigma_k \in [\sigma_{\min}, \sigma_{\max}]. \tag{17}$$

$$\mathcal{L}_{\mathrm{NLL}} = -\frac{1}{|\Omega|} \sum_{(q,t)\in\Omega} \log \sum_{k} \exp\Big(\log \pi_k - \tfrac{1}{2}\Big(\tfrac{y-\mu_k}{\sigma_k}\Big)^2 - \log \sigma_k - \tfrac{1}{2}\log 2\pi \Big), \tag{18}$$

where Ω is the set of target positions with an actual observation.

7.3 CRPS via the order-statistic identity

NLL alone is unbounded and unstable early in training. We add the continuous ranked probability score, estimated from n samples $x_1, \dots, x_n$:

$$\mathrm{CRPS} \ \approx\ \frac{1}{n}\sum_{i} |x_i - y| \ -\ \frac{1}{2n(n-1)} \sum_{i \neq j} |x_i - x_j|. \tag{19}$$

The naive pairwise term materialises an $(n, n, \dots)$ tensor and will exhaust memory at realistic query counts. Using the order statistics $x_{(1)} \leq \cdots \leq x_{(n)}$,

$$\boxed{\sum_{i,j} |x_i - x_j| \;=\; 2 \sum_{i=1}^{n} \big(2i - n - 1\big)\, x_{(i)}} \tag{20}$$

which reduces the cost to $O(n \log n)$ time and $O(n)$ memory. *This is not an optimisation detail — the naive form is what makes the loss un-runnable at scale.*

7.4 Total objective

$$\mathcal{L} = w_{\text{nll}}\mathcal{L}_{\text{NLL}} \;+\; w_{\text{crps}}\mathcal{L}_{\text{CRPS}} \;+\; w_{\text{phys}}\mathcal{L}^{\text{recon}}_{\text{CRPS}}, \tag{21}$$

where the third term scores the *reconstructed* observables ($PM_{2.5}$, NO, NO_2) obtained by transforming the sampled ensemble, so the constraint is scored as a distribution rather than as a point identity. Optionally weight targets by $1 + \alpha\, \sigma\big(4(y - y_{\text{thr}})\big)$ to keep winter episodes from being averaged into irrelevance by clean summer months.

8 Pre-training curriculum

8.1 Mixed masking

Random patch masking alone yields a model that is excellent at imputation and mediocre at forecasting, because there is always future context on both sides of every gap. The mask distribution is therefore a mixture:

Regime	Weight	Construction
Tail	0.50	Mask the last k patches, $k \sim \mathcal{U}[1, H]$. No future context: this is forecasting.
Block	0.35	Station-wide outages, all channels, durations $L \sim \text{LogNormal}(\hat{\mu}, \hat{\sigma})$ fitted to the real gap distribution.
Channel	0.15	Drop a random channel subset over a random window: forces $PM_{2.5}$ to be forecast from meteorology and neighbours alone.

Table 4: Masking mixture. The block regime is station-wide precisely because the EDA establishes systemic rather than independent failure.

Crucially, $\hat{\mu}, \hat{\sigma}$ are *estimated from the data*, not guessed: the observed run-length distribution of zeros in M is exactly the outage distribution the model must learn to survive.

8.2 Two-stage schedule

1. **Stage 1 — national masked pre-training.** All-India corpus, mixed masking, AdamW ($\beta = (0.9, 0.95)$, weight decay 0.05), cosine schedule with linear warmup, gradient clipping at 1.0.

2. **Stage 2 — horizon fine-tuning.** Delhi corpus, strict causal split, tail masking only, fine-tuned at the operational horizons (1 h, 6 h, 24 h, 72 h) with lower learning rate.

Group	Parameter	Value	Note
Window	context	512 steps (5.3 d)	at 15-minute cadence
	patch P	16 (4 h)	$N = 32$ context patches
	horizon	288 steps (72 h)	$H = 18$ horizon patches
	neighbourhood	$k = 12$ nearest stations	per training sample
Encoder	d_{model}	384	
	heads	8	
	d_{ff}	1536	
	axial blocks	4	= 12 attention layers
	latents/station	16	Perceiver bottleneck
Decoder	layers	4	self + cross + FFN
	mixture components K	3	log-normal
Training	optimiser	AdamW, lr 3×10^{-4}	wd 0.05, clip 1.0
	CRPS samples	32	order-statistic estimator
	normalisation	masked RevIN	context statistics only
Size	parameters (Delhi)	$\approx$ 23M	measured
	parameters (national)	$\approx$ 150M	$d = 768$, 8 blocks

Table 5: Reference configuration. Activation memory, not FLOPs, is the binding constraint: the axial encoder stores ≈ 1.5 GB of activations per sample at this size, so gradient checkpointing over the encoder blocks (roughly 4× memory reduction for $\sim$30% step-time cost) is what makes a useful batch size fit on a single card.

9 Reference configuration

10 Evaluation protocol

10.1 Metrics that matter

Aggregate MAE over a full year is dominated by clean summer months and will look excellent for a model that is useless in December. Report:

1. **Skill stratified by AQI band**, not pooled.
2. **Episode recall and precision**: detection of $\text{PM}_{2.5} > 250\,\mu\text{g/m}^3$ at 24-hour and 48-hour lead. This is the number that decides whether the model is operationally useful.
3. **Calibration**: PIT histograms and reliability of the 90% and 95% predictive intervals, per horizon.
4. **CRPS by lead time**, to expose where skill decays to climatology.
5. **Imputation and forecasting scored separately.** They are different tasks and a single pooled number hides failure in one of them.

10.2 Baselines

10.3 Splits and leakage discipline

Hold out 2025–2026 entirely; use rolling-origin backtests within the training period. Two leakage paths are easy to create and hard to notice:

- Normalisation statistics computed over the full window rather than the context. Test this explicitly by perturbing the horizon and asserting the encoder output is bit-identical.

Baseline	Why it is included
Persistence	Floor; anything below this is broken
Seasonal-naive (same hour yesterday)	Captures the diurnal cycle for free
Climatology by (station, hour, month)	The honest long-horizon competitor
Gradient-boosted trees on lagged features	**Brutally strong on this task**
PatchTST / iTransformer, channel-independent	Isolates the value of spatial attention
Design A (per-pollutant decoders)	Isolates the value of the query decoder

Table 6: If the foundation model does not clearly beat gradient-boosted trees at 24 h+, the problem is in the setup, not in the baseline.

- Masked ground truth reaching the model through the mask channel. Assert that the input mask and target mask never overlap.

11 Planned ablations

Ablation	Removes	Hypothesis
No spatial attention	Cross-station information	Large loss during block outages
Distance-only bias	Wind asymmetry	Moderate loss at 6–24 h lead
Design A decoders	Query decoder	Coherence violations; no skill gain
Random-only masking	Tail regime	Imputation holds, forecasting degrades
Point head + MSE	Distributional output	Episode recall collapses
No physical reparameterisation	Constraints	$PM_{2.5}$ >PM_{10} violations appear
Delhi-only pre-training	National corpus	Quantifies the transfer benefit
No boundary-layer covariate	ERA5 PBLH	Expected largest single-feature effect

12 Risks and honest caveats

Scale.
2.16×10^8 observations is a well-sized deep learning dataset, not a foundation-model corpus. The "foundation model" claim rests on national pre-training plus demonstrated transfer, and should be stated that way.

Scaling-law transfer.
The ~ 20 tokens/parameter heuristic is borrowed from language modelling and is not established for time series. Treat the parameter-count guidance as an order-of-magnitude prior, and run a small scaling sweep before committing to the national model size.

NWP dependence.
Training with ERA5 as a perfect-prognosis oracle overstates operational skill. The oracle-versus-forecast gap must be reported.

Non-stationarity.
Emission inventories, GRAP interventions and the vehicle fleet all change across 2018–2026. A model evaluated only on a held-out tail may be measuring drift rather than skill; include a rolling re-fit comparison.

Station representativeness.
CPCB station siting is not random; a kerbside monitor and a background monitor are different measurement processes. Static covariates U should include a site-type indicator.

13 Implementation map

The accompanying reference implementation is organised as follows.

File	Contents
`aqfm/config.py`	`AQFMConfig`; `.small()` for tests
`aqfm/layers.py`	masked RevIN, Fourier time, masked attention, axial block, wind bias
`aqfm/model.py`	tokeniser, encoder, latent pooler, query decoder, mixture head
`aqfm/losses.py`	mixture NLL, order-statistic CRPS, episode weighting
`aqfm/masking.py`	mask sampler, empirical outage-length fitting
`aqfm/physical.py`	ratio channels and constrained reconstruction
`aqfm/data.py`	windowing, neighbourhood sampling, query construction
`aqfm/train.py`	two-stage training, evaluation, baselines
`tests/`	shape, leakage, masking, constraint and loss-ordering tests

Listing 1: Minimal usage.

```
from aqfm import AQFMConfig, AQFM, AirQualityWindows
from aqfm.train import train, evaluate, baselines

cfg = AQFMConfig(grad_checkpoint=True)
ds  = AirQualityWindows(values, observed, timestamps_h, coords, static,
                        cfg, n_neighbours=12, pretrain=True)   # stage 1
model = train(cfg, ds, steps=200_000, batch_size=16, device="cuda")

ds_ft = AirQualityWindows(..., pretrain=False)                 # stage 2
model = train(cfg, ds_ft, model=model, steps=20_000, lr=5e-5)

print(evaluate(model, ds_ft, cfg), baselines(ds_ft, cfg))
```